Numerical Integration.

Trapezoidal Rule

$$I = \int_a^b F(x)\,dx$$

So, This is only use 2 subIntervals.

$$I = \frac{b-a}{2n}\left(F(b)+F(a)\right)$$

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IF we have n intervals.

$$I = \frac{b-a}{2n}\left\{F(x_0) + \sum_{i=1}^{n-1} 2F(x_i) + F(x_n)\right\}$$

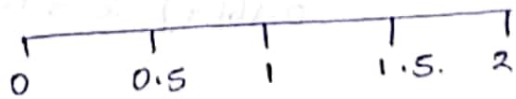
Ex:-

$F(x) = x^2$

$a = 0$, $b = 2$, $n = 4$

$$h = \frac{2-0}{4} = 0.5$$

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$F(0) = 0$

$F(0.5) = 0.25$

$F(1) = 1$

$F(1.5) = 2.25$

$F(2) = 4$

$$I = \frac{2-0}{4\times 2}\left\{0 + 2(0.25+1+2.25)+4\right\}$$

$$= \frac{0.5}{2}\left\{2(3.5)+4\right\}$$

$$= \frac{0.5}{2} \times 11$$

$$= \frac{5.5}{2} = 2.75$$

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Simpson's 1/3 rule

$$\int_a^b F(x)\,dx = \frac{h}{3}\left[F(x_0) + 2\sum_{i=2,\,inc\,2}^{n-2} F(x_i) + 4\sum_{i=1,\,inc\,2}^{n-1} F(x_i) + F(x_n)\right]$$

Even odd

Ex:- $F(x) = x^2$.

$a = 0 \quad b = 2 \quad n = 4$

$h = 0.5$

$x_0 = 0 \quad x_1 = 0.5 \quad x_2 = 1 \quad x_3 = 1.5 \quad x_4 = 2.$

$F(x_0) = 0 \quad F(x_1) = 0.25 \quad F(x_2) = 1 \quad F(x_3) = 2.25 \quad F(x_4) = 4$

$$I = \frac{0.5}{3}\left[F(x_0) + 2\sum_{i=2}^{2} F(x_i) + 4\sum_{i=1}^{3} F(x_i) + F(x_4)\right]$$

$$= 0.1667\left[0 + 2(1) + 4[0.25 + 2.25] + 4\right]$$

$$= 0.1667\left[2 + 14\right]$$

$$= 2.667.$$

Simpson's 3/8 Rule

$$I = \frac{3h}{8}\left[F(x_0) + 3F(x_1) + 3F(x_2) + 2F(x_3) + F(x_4)\right]$$

$$= \frac{3h}{8}\left[F(x_0) + 3\{F(x_1) + F(x_2) + F(x_4) + F(x_5) + 2(F(x_3) + F(x_6))\} + F(x_7)\right]$$